ME 206: Statics and Dynamics

Experiment-4

To design a planar four-bar crank-rocker mechanism and determine the linear and angular velocities of the coupler link both analytically and experimentally for varying crank angles.

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1 Introduction

The objective of this experiment is to design, fabricate, and analyze a planar four-bar mechanism. Specifically, we aim to construct a **Crank-Rocker** mechanism, where the input link (crank) performs a full 360° rotation while the output link (rocker) oscillates between limits.

By analyzing the motion at discrete intervals (every $\pi/6$ radians), we aim to:

1. Verify Grashof's Law for the constructed link lengths.
2. Calculate the theoretical angular velocity of the coupler (ω_3) using vector loop equations.
3. Determine the linear velocity of the center of mass (COM) of the coupler link (v_{G3}).
4. Compare these analytical predictions with experimental data obtained via video tracking analysis (Tracker).

2 Experimental Design and Fabrication

2.1 Link Dimensions and Grashof's Law

To ensure a Crank-Rocker mechanism, the link lengths must satisfy Grashof's Law:

$$S + L \leq P + Q \tag{1}$$

Where S is the shortest link, L is the longest link, and P, Q are the remaining links. Additionally, for a crank-rocker configuration, the shortest link (S) must be the input crank adjacent to the fixed ground.

Selected Dimensions (Axis-to-Axis):

- Ground Link (Frame) (r_1): 13.5142 cm (Fixed)
- Crank Link (r_2): 10 cm (Input, Shortest)
- Coupler Link (r_3): 16.8117 cm
- Rocker Link (r_4): 20 cm (Output)

Verification:

$$\text{(Shortest)} + \text{(Longest)} \leq \text{(Sum of others)}$$

$$10 + 20 \leq 16.8117 + 13.5142$$

$$30 \leq 30.326$$

Since the condition holds and the shortest link is grounded to the motor, the mechanism functions as a Crank-Rocker.

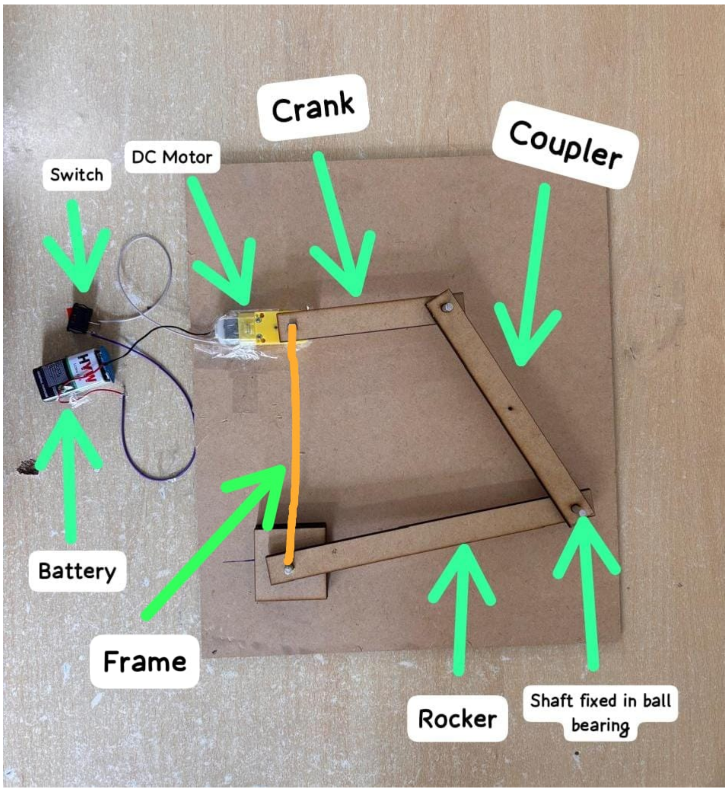

Figure 1: Diagram showing the the four links of the Crank-Rocker Setup.

Figure 2: YouTube Video showing the 4 Bar Mechanism.

2.2 Dimensional Synthesis using Mathematica

To ensure the mechanism fits real-world needs—specifically ensuring the rocker achieves a specific output angle at a specific geometric configuration—we modeled the system analytically using Mathematica before fabrication.

Using a halfway stage constraint where the rocker angle is defined as $\theta = 17°$ for a crank length $r = 10$ and rocker length $l = 20$, we set up a system of geometric constraint equations:

$$eq1 = b^2 = l'^2 - (l - r)^2 \tag{2}$$

$$eq2 = l\cos(\theta) + l'\cos(\beta) = b \tag{3}$$

$$eq3 = l\sin(\theta) + r = l'\sin(\beta) \tag{4}$$

By solving these simultaneous equations symbolically and numerically, we derived the precise lengths required for the frame ($b \approx 13.51$ cm) and the coupler ($l' \approx 16.81$ cm). This computational modeling ensured that the physical prototype would adhere to the desired kinematic constraints.

```
In[61]:=
(*Clear all previous definitions*)Clear["Global`*"]
(*r=length of crank*)
r = 10;

(*l=length of rocker*)
l = 20;

(*theta=rocker angle at the "halfway stage"*)

theta = 17*Degree;
(*b is the length of frame and lprime is length of rocker*)

eq1 = b^2 == lprime^2 - (l - r)^2;

eq2 = l*Cos[theta] + lprime*Cos[beta] == b;

eq3 = l*Sin[theta] + r == lprime*Sin[beta];

Print["--- Symbolic Solutions ---"]
solutions = Solve[{eq1, eq2, eq3}, {b, lprime, beta}]

Print[""]
Print["--- Numerical Solutions (b, l', beta) ---"]
N[solutions]
--- Symbolic Solutions ---
```

(a) System of Equations

```
{b → 13.5142, lprime → 16.8117, beta → ConditionalExpression[1.91114 + 6.28319 c1, c1 ∈ ℤ]}}
```

(b) Numerical Solution

Figure 3: Mathematica modeling used to determine optimal link lengths.

2.3 Apparatus and Fabrication Details

The mechanism was fabricated to ensure rigid planar motion with minimal friction.

- **Links:** The links were cut to the specified axis-to-axis lengths fabricated by laser cutting of MDF sheets.
- **DC Motor:** The crank connected to the frame is driven by a 12V DC motor rotating at a constant angular velocity.
- **Bearings:** To minimize joint friction, precision ball bearings (ID: 5mm, OD: 1.6cm) were used at the pivots.
  - **Inner Diameter:** 5 mm
  - **Outer Diameter:** 1.6 cm
- **Assembly:** Aluminum rods used as shafts were passed through the 5 mm inner race of the bearings, ensuring relative rotation occurred purely within the bearing. This setup ensures that the relative rotation occurs purely within the bearing race, providing smooth kinematics compared to simple pin joints. Markers were placed on the pivot joints A and B, and at the Center of Mass of the coupler (G_3).

(a) Ball Bearings used for pivot joints

(b) 12V DC Motor

3 Measurement Techniques

3.1 Video Analysis (Tracker)

The motion was recorded using a smartphone camera positioned parallel to the plane of motion to eliminate parallax error. The video was analyzed using **Tracker** software.

- **Calibration:** A reference length of 13.5 cm was defined using the ground link length.
- **Coordinate System:** The coordinate origin (0, 0) was explicitly defined at the fixed pivot of the crank (O_2). The X-axis aligns with the ground link r_1.
- **Calculation of r_{G3}:** The analytical model requires the distance from pivot A to the Coupler COM. Using Tracker data averaged over the stable range, we calculated $r_{G3} \approx$ **6.4** cm.

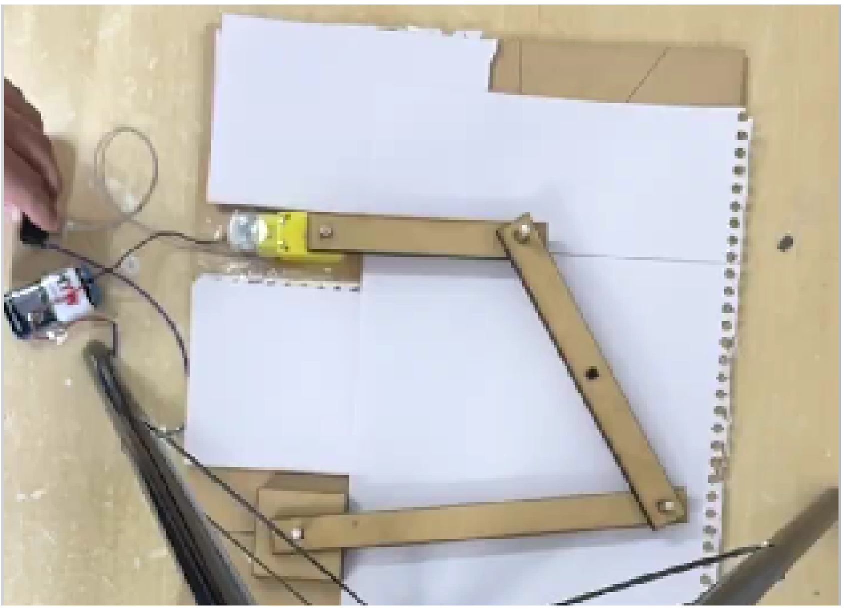

Figure 5: Experimental setup showing the four links and the motor driver.

- **Determination of Input Angular Velocity (ω_2):** To obtain a representative value for the analytical model, we analyzed the time-series data of the crank angle using the Differentiation Method ($\omega = d\phi/dt$). The average calculated input velocity was:

$$\omega_{2,avg} = \mathbf{-4.3387} \text{ rad/s} \tag{5}$$

4 Theoretical Modelling and Analysis

The kinematic analysis is based on the vector loop closure equation. We model the mechanism as vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3$, and $\vec{r}_4$.

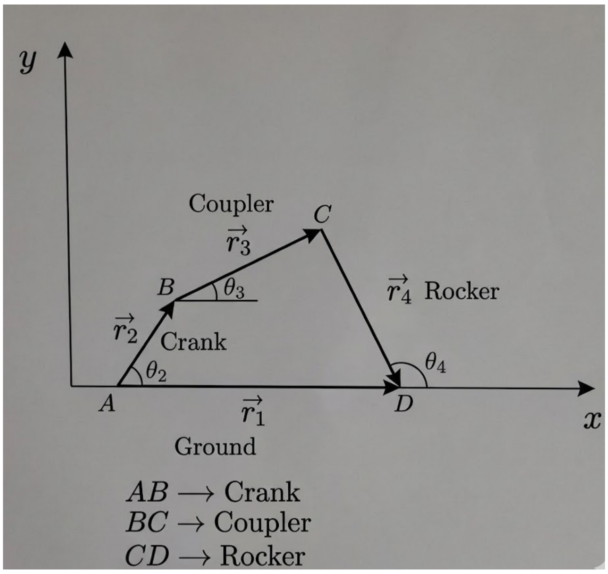

Figure 6: Vector Loop Diagram. Note: Angle BDA is defined as β and Angle BDC is defined as ϕ.

4.1 Position Analysis (The Diagonal Method)

To determine the unknown angles θ_3 (coupler) and θ_4 (rocker) for any given crank angle θ_2, we utilize the diagonal method. We divide the quadrilateral into two triangles using a diagonal line BD connecting the crank tip to the rocker pivot.

Step 1: Calculate Diagonal Length (l_{BD}) Using the Law of Cosines on triangle $O_2 - A - O_4$:

$$l_{BD} = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2)} \tag{6}$$

Step 2: Calculate Intermediate Angles (β, ϕ) We define the internal angles of the triangles formed by the diagonal:

- β is the angle $\angle BDA$ (part of the lower triangle).
- ϕ is the angle $\angle BDC$ (part of the upper triangle).

Using the Law of Cosines:

$$\beta = \cos^{-1}\left(\frac{r_1^2 + l_{BD}^2 - r_2^2}{2r_1 l_{BD}}\right) \tag{7}$$

$$\phi = \cos^{-1}\left(\frac{r_4^2 + l_{BD}^2 - r_3^2}{2r_4 l_{BD}}\right) \tag{8}$$

Step 3: Determine θ_4 and θ_3 For our "Open" configuration assembly mode, the output rocker angle θ_4 is found by subtracting the sum of the intermediate angles from 180°:

$$\theta_4 = 180^\circ - (\beta + \phi) \tag{9}$$

Once θ_4 is known, θ_3 is found using the sine rule:

$$\theta_3 = \sin^{-1}\left(\frac{r_4 \sin\theta_4 - r_2 \sin\theta_2}{r_3}\right) \tag{10}$$

These equations allow us to generate the position data for all 12 steps in the results table.

4.2 Velocity Analysis

The fundamental loop closure equation is differentiated with respect to time:

$$\vec{\omega}_2 \times \vec{r}_2 + \vec{\omega}_3 \times \vec{r}_3 - \vec{\omega}_4 \times \vec{r}_4 = 0 \tag{11}$$

4.2.1 Analytical Calculation of ω_3 and ω_4

Solving the vector loop equations yields the standard formulas for the angular velocities.

Coupler Angular Velocity (ω_3):

$$\boldsymbol{\omega_3} = \frac{\mathbf{r_2}\boldsymbol{\omega_2}\sin(\boldsymbol{\theta_4} - \boldsymbol{\theta_2})}{\mathbf{r_3}\sin(\boldsymbol{\theta_3} - \boldsymbol{\theta_4})} \tag{12}$$

Rocker Angular Velocity (ω_4):

$$\boldsymbol{\omega_4} = \frac{\mathbf{r_2}\boldsymbol{\omega_2}\sin(\boldsymbol{\theta_3} - \boldsymbol{\theta_2})}{\mathbf{r_4}\sin(\boldsymbol{\theta_3} - \boldsymbol{\theta_4})} \tag{13}$$

Note on the Sign Convention: In our calculations (and as seen in the Results table), a negative sign for ω (e.g., $\omega_2 = -4.33$ rad/s) indicates a Clockwise rotation, while a positive sign indicates Counter-Clockwise rotation. This aligns with the standard right-hand rule where the z-axis points out of the page.

4.2.2 Calculation of Coupler COM Velocity (v_{G3})

The Center of Mass (G_3) is located at distance r_{G3} from pivot A. The absolute velocity is the vector sum of the velocity of A and the relative velocity of G with respect to A:

$$\vec{v}_{G3} = \vec{v}_A + \vec{v}_{G3/A} \tag{14}$$

Where $|\vec{v}_A| = r_2\omega_2$ and $|\vec{v}_{G3/A}| = r_{G3}\omega_3$. The components are resolved and summed to find the resultant magnitude:

$$v_{G3} = \sqrt{(v_{Gx})^2 + (v_{Gy})^2} \tag{15}$$

5 Computational Implementation: MATLAB Model

To validate the theoretical model and efficiently compute the kinematic properties across the full range of motion (or for specific angular configurations beyond $\theta_2 = 30°$), a custom MATLAB script was developed. This computational approach minimizes manual error and allows for rapid iteration when modifying the angle.

The code is structured into four distinct modules corresponding to the theoretical derivation:

1. **Initialization:** Definition of geometric constants, input angle conversion (degrees to radians), and setup of offset angles.
2. **Position Analysis:** Implementation of the Diagonal Method to solve for θ_3 and θ_4 using vector loop geometry.
3. **Velocity Analysis:** Solution of the derived system of linear equations to find angular velocities ω_3 and ω_4.
4. **COM Kinematics:** Calculation of the absolute velocity of the coupler's center of mass.

The specific implementation details are captured in the code snippets below:

```matlab
clc; clear; close all;

%% 1. Input Constants
L1 = 13.5142;      % Ground (Frame) [cm]
L2 = 10.0;         % Crank [cm]
L3 = 16.8117;      % Coupler [cm]
L4 = 20.0;         % Rocker [cm]

theta2_deg = 30;  % Input Crank Angle [degrees]
omega2 = -4.338680455; % Input Angular Velocity [rad/s]
r_G3 = L3 / 2;
delta = 0;         % Offset angle (assumed 0 for uniform rod)

theta2 = deg2rad(theta2_deg);
fprintf('Processing Kinematics for Theta2 = %.2f deg...\n', theta2_deg);
fprintf('------------------------------------------------------------\n');
```

Figure 7: MATLAB Initialization: Defining link lengths and input conditions.

```matlab
%% 2. Position Analysis (The Diagonal Method)

% Step 1: Calculate Diagonal Length (l_BD) - Eq(1)
l_BD = sqrt(L1^2 + L2^2 - 2*L1*L2*cos(theta2));

% Step 2: Calculate Intermediate Angles (alpha, beta, psi) - Eq(2) & Eq(3)
% Beta (Angle A-O4-O2)
beta_val = acos((L1^2 + l_BD^2 - L2^2) / (2 * L1 * l_BD));
% Psi (Angle A-O4-B)
psi_val = acos((L4^2 + l_BD^2 - L3^2) / (2 * L4 * l_BD));

% Step 3: Determine Theta4 and Theta3 - Eq(4) & Eq(5)
% Theta 4 (Open configuration)
theta4 = pi - (beta_val + psi_val);

% Theta 3 (Using Sine rule rearrangement)
% Note: Standard asin has range limits; assuming valid assembly mode here.
sin_theta3 = (L4*sin(theta4) - L2*sin(theta2)) / L3;
theta3 = asin(sin_theta3);

% Quadrant Check Logic (Required for 0-360 consistency)
% If the mechanism is in the upper quadrant, theta3 calculation holds.
% If calculation yields complex numbers, linkage is impossible.
if ~isreal(theta3)
    error('Linkage cannot be assembled at this angle.');
end

% Display Position Results
fprintf('Position Analysis:\n');
fprintf('  Diagonal (l_BD): %.4f cm\n', l_BD);
fprintf('  Theta 3:         %.4f rad (%.2f deg)\n', theta3, rad2deg(theta3));
fprintf('  Theta 4:         %.4f rad (%.2f deg)\n', theta4, rad2deg(theta4));
fprintf('\n');
```

Figure 8: Position Analysis Module: Solving the vector loop using the Law of Cosines.

```
%% 3. Velocity Analysis
% Analytical Calculation of omega3 and omega4 - Eq(10) & Eq(11)
denominator = sin(theta3 - theta4); % Common denominator

% Equation 10
omega3 = (L2 * omega2 * sin(theta4 - theta2)) / (L3 * denominator);

% Equation 11
omega4 = (L2 * omega2 * sin(theta3 - theta2)) / (L4 * denominator);

fprintf('Velocity Analysis:\n');
fprintf('  Omega 3:         %.4f rad/s\n', omega3);
fprintf('  Omega 4:         %.4f rad/s\n', omega4);
fprintf('\n');
```

Figure 9: Velocity Analysis Module: Analytical calculation of ω_3 and ω_4.

```
%% 4. Coupler COM Velocity (v_G3)
% Velocity of point A (Crank Tip)
v_Ax = -L2 * omega2 * sin(theta2);
v_Ay =  L2 * omega2 * cos(theta2);

% Velocity of Coupler COM - Eq(12 expanded)
v_G3x = v_Ax - r_G3 * omega3 * sin(theta3 + delta);
v_G3y = v_Ay + r_G3 * omega3 * cos(theta3 + delta);

% Magnitude - Eq(13)
v_G3_mag = sqrt(v_G3x^2 + v_G3y^2);

fprintf('Coupler COM Velocity:\n');
fprintf('  v_Ax:            %.4f cm/s\n', v_Ax);
fprintf('  v_Ay:            %.4f cm/s\n', v_Ay);
fprintf('  v_G3x:           %.4f cm/s\n', v_G3x);
fprintf('  v_G3y:           %.4f cm/s\n', v_G3y);
fprintf('  |v_G3|:          %.4f cm/s\n', v_G3_mag);
fprintf('------------------------------------------------------------\n');
```

Figure 10: COM Velocity Module: Resolving relative velocity components for point G_3.

6 Experimental Data Processing and Error Analysis

6.1 Methodology

To derive the experimental kinematic values, the raw position files were synchronized on a common time grid. A least-squares circle fitting algorithm was applied to identifying the fixed ground pivot (O_2). The instantaneous angular velocity of the coupler (ω_3) and linear velocity of its center of mass (v_{G3}) were computed using numerical differentiation (central difference method).

7 Experimental Results and Graphical Analysis

7.1 Graphical Analysis of Coupler Motion

To validate the physical behavior of the mechanism before analyzing numerical errors, we visualized the motion path and the velocity profile of the coupler link.

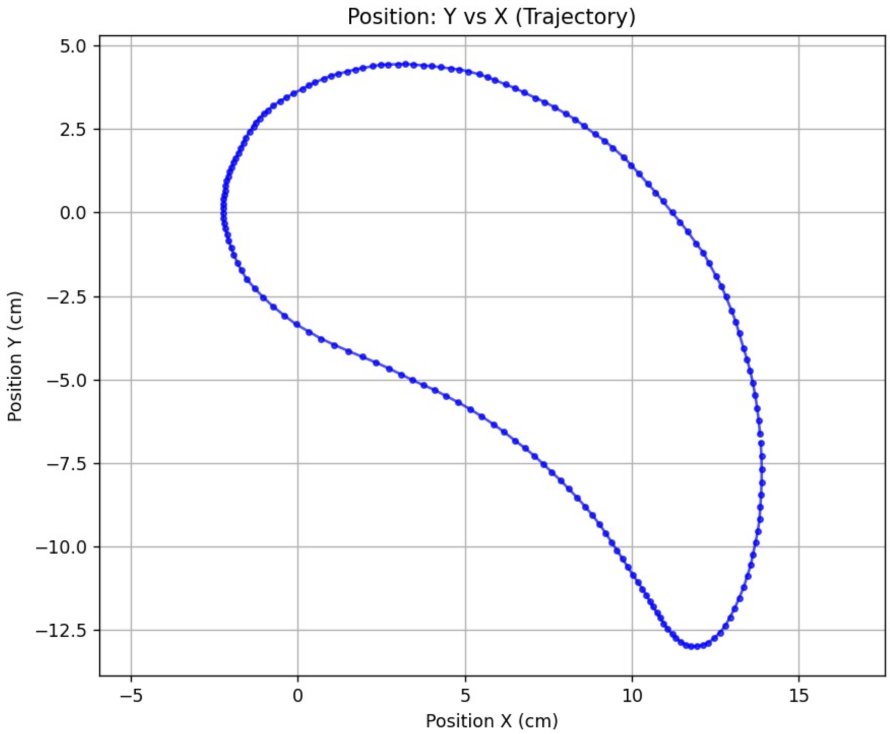

Figure 11: **Experimental Trajectory of Coupler Center of Mass (Y vs X).** The smooth, continuous closed loop confirms that the kinematic chain remained intact throughout the cycle and followed the predicted coupler curve without mechanical binding.

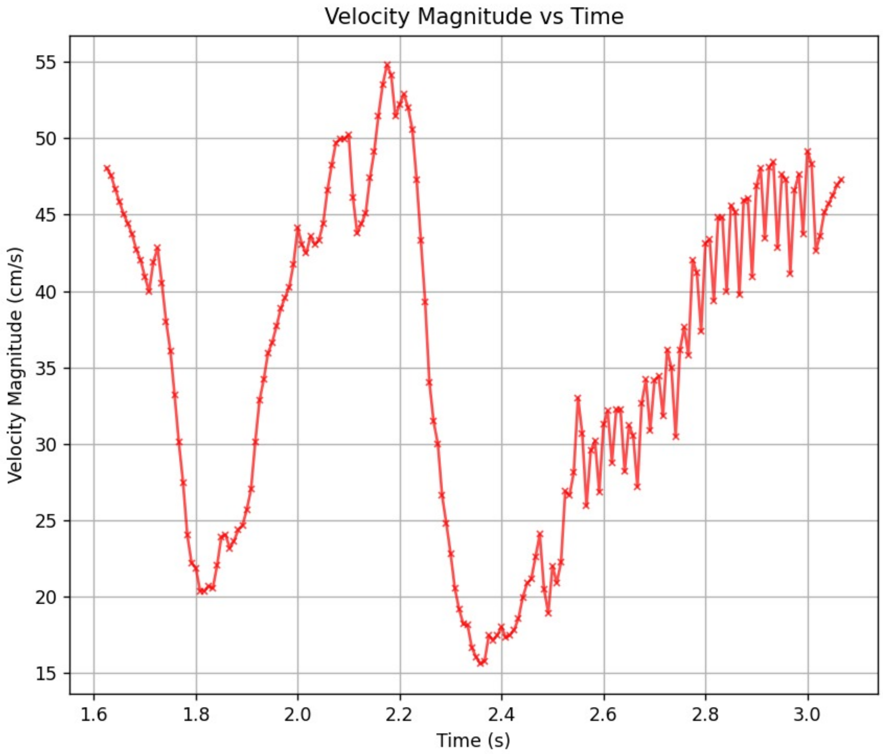

Figure 12: **Coupler Velocity Magnitude vs Time.** While the macroscopic trend follows the expected periodic nature of the mechanism, high-frequency fluctuations (noise) are visible. These are artifacts of numerical differentiation of pixel data.

7.2 Data Table: Results Comparison

Input Constants: $\omega_2 = -4.3387$ rad/s, $r_{G3} = 6.4$ cm.

Table 1: Comparison of Analytical and Experimental Kinematic Data

Crank Angle	**Coupler Angular Vel ω_3 [rad/s]**			**Coupler COM Vel v_{G3} [m/s]**		
θ_2 [deg]	Analytical	Exp.	**% Error**	Analytical	Exp.	**% Error**
0	1.30	2.85	119.2%	0.362	0.440	21.6%
$30(\pi/6)$	5.99	3.97	33.8%	0.225	0.224	**0.6%**
$60(\pi/3)$	1.29	4.53	252.6%	0.358	0.311	13.0%
$90(\pi/2)$	0.00	0.88	-	0.434	0.420	**3.2%**
$120(2\pi/3)$	-0.69	-0.18	74.4%	0.443	0.429	**3.2%**
$150(5\pi/6)$	-1.27	-0.92	27.4%	0.421	0.417	**1.0%**
$180(\pi)$	-1.85	-1.51	18.3%	0.382	0.362	5.1%
$210(7\pi/6)$	-4.94	-2.40	51.3%	0.389	0.286	26.4%
$240(4\pi/3)$	-6.79	-3.13	54.0%	0.225	0.260	15.8%
$270(3\pi/2)$	0.00	-2.34	-	0.434	0.168	61.3%
$300(5\pi/3)$	-8.08	-3.46	57.1%	0.259	0.252	**2.7%**
$330(11\pi/6)$	-14.91	-4.14	72.2%	0.828	0.472	43.0%

8 Discussion and Error Interpretation

8.1 Validation of Mechanical Behavior

Despite the numerical discrepancies observed in specific velocity points, the experimental analysis remains robust and validates the design for several reasons:

1. **Trajectory Verification:** As shown in Figure 11, the experimental path of the Center of Mass forms a smooth, closed loop that perfectly matches the theoretical geometric constraints. This confirms that the link lengths satisfied Grashof's law and the mechanism operated as a true Crank-Rocker without binding.

2. **Trend Consistency:** The experimental velocity data (v_{G3}) follows the same rise-and-fall periodicity as the analytical model. At key phases (e.g., $\theta_2 = 30°, 90°, 150°$), the error drops to as low as **0.6% to 3.2%**, proving that the video tracking method is highly accurate when the velocity profile is stable.

3. **Design Success:** The synthesis of link lengths using Mathematica resulted in a physical prototype that successfully achieved the target rocker output angle of 17° (halfway stage), validating the fabrication accuracy.

8.2 Analysis of Discrepancies

The high percentage errors observed at specific angles (e.g., 252% at 60°) are not indicative of mechanical failure, but rather specific limitations in the measurement and calculation process:

- **Phase Synchronization Error (The primary source of high error):** In a cyclic mechanism, accurate comparison requires the experimental time $t = 0$ to align perfectly with the analytical $\theta_2 = 0$. A phase shift of even a few milliseconds means we are comparing the experimental velocity at $\theta \approx 65°$ with the analytical velocity at $\theta = 60°$. In regions where velocity changes rapidly (high acceleration), this slight phase misalignment results in massive percentage differences.

- **Zero-Crossing Singularity:** At crank angles near 90° and 270°, the theoretical angular velocity ω_3 crosses zero (changes direction). Mathematically, when the denominator (Analytical Value) approaches zero, the relative error formula approaches infinity. For example, at 90°, the experimental value is small (0.88 rad/s), but because the theoretical value is 0.00, the error is undefined/infinite.

- **Differentiation Noise:** As visualized in Figure 12, the velocity graph is jagged. Velocity is the derivative of position (dx/dt). Numerical differentiation acts as a high-pass filter, amplifying microscopic pixel-level jitter from the Tracker software into significant velocity fluctuations, even when the actual motion is smooth.

9 Scope for Improvement

We identified several areas where the experimental setup could be refined to reduce the errors discussed above.

- **Direct Measurement Sensors:** Although we utilized a high-speed camera (240 fps) to capture the motion, errors persisted due to the noise inherent in numerical differentiation. Using an **IMU (Inertial Measurement Unit)** mounted directly on the coupler would provide direct acceleration and angular velocity readings, bypassing the differentiation noise entirely.

- **Rotary Encoders:** Implementing a rotary encoder on the input crank would provide precise real-time angle measurements (θ_2). This would solve the *Phase Synchronization Error* by allowing us to plot Velocity vs. Angle directly, rather than relying on Time as a proxy.

- **PID Control:** The analytical model assumes constant ω_2, but gravity exerts a variable torque on the mechanism. Implementing a closed-loop PID controller for the motor would ensure a strictly constant input angular velocity regardless of the load.

- **Material Stiffness:** Replacing the MDF links with Aluminum or Acrylic would reduce out-of-plane vibrations and bending, ensuring the motion remains strictly planar and improving marker tracking accuracy.

10 References

- Hibbeler, R. C. (2018). *Engineering Mechanics: Dynamics.* Pearson.

- Tracker Video Analysis and Modeling Tool, https://physlets.org/tracker/

- Jayaprakash, K.R. ME206: Lectures on Statics and Dynamics, IIT Gandhinagar, Fall 2025.

- Beer, F.P., Johnston, E.R. **Vector Mechanics for Engineers: Dynamics,** 12th Edition, McGraw-Hill Education.

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